Tp_priority.v

Source code

```
`timescale 1ns / 1ps

module tb_priority();

    reg [3:0] r_a;
    wire [1:0] w_led;
    wire w_v;

    priority_encoder u_priority(
        .a(r_a),
        .led(w_led),
        .v(w_v)
    );

    initial begin
        r_a = 4'b0000;
        #10;

        $monitor("a = %b | led = %b | v = %b",r_a,w_led,w_v);

        r_a = 4'b0001;
        #10;

        r_a = 4'b0010;
        #10;

        r_a = 4'b0011;
        #10;

        r_a = 4'b0100;
        #10;

        r_a = 4'b0101;
        #10;

        r_a = 4'b0110;
        #10;

        r_a = 4'b0111;
        #10;

        r_a = 4'b1000;
```

```
        #10

        r_a = 4'b1001;
        #10;

        r_a = 4'b1010;
        #10;

        r_a = 4'b1011;
        #10;

        r_a = 4'b1100;
        #10;

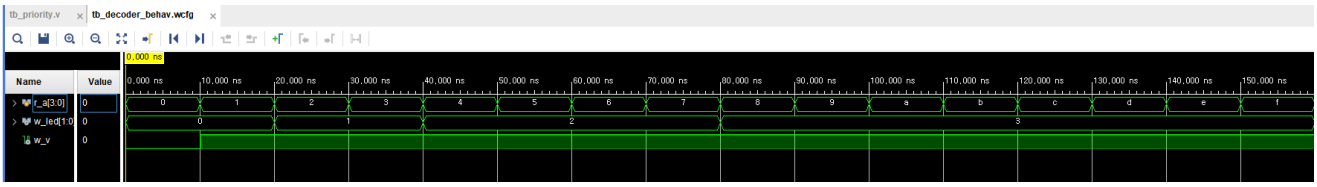
        r_a = 4'b1101;
        #10;

        r_a = 4'b1110;
        #10;

        r_a = 4'b1111;
        #10;

        $finish;
    end
endmodule
```

시뮬레이션 파형



Priority_encoder.v

Source code

```
`timescale 1ns / 1ps

module priority_encoder(
```

```
    input [3:0] a,
    output [1:0] led,
    output v
    );

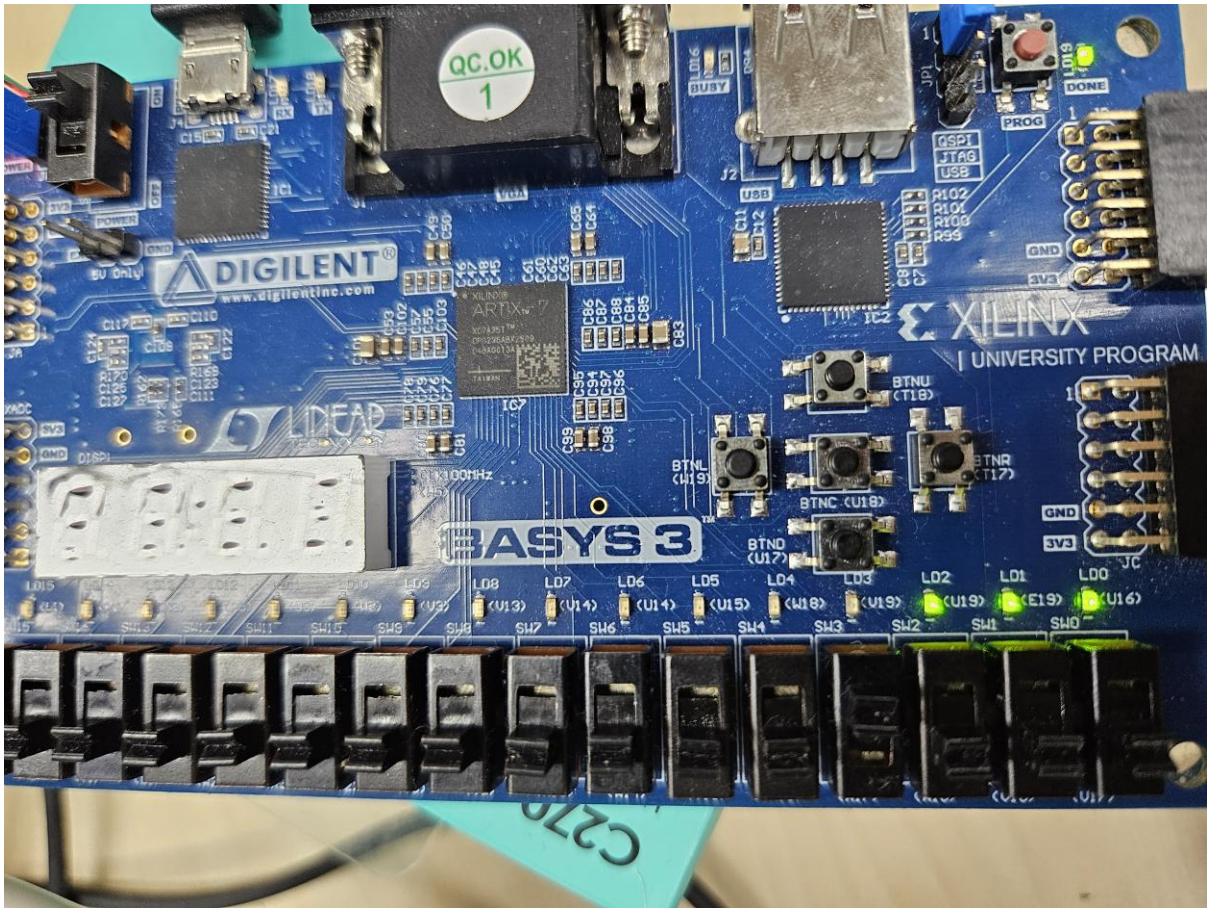
    reg [1:0] r_led;
    reg r_v;

    always @(*) begin

        if (a[3]) begin
            r_led = 2'b11;
            r_v = 1'b1;
        end
        else if (a[2]) begin
            r_led = 2'b10;
            r_v = 1'b1;
        end
        else if (a[1]) begin
            r_led = 2'b01;
            r_v = 1'b1;
        end
        else if (a[0]) begin
            r_led = 2'b00;
            r_v = 1'b1;
        end
        else begin
            r_led = 2'b00;
            r_v   = 1'b0;
        end
    end

    assign led = r_led;
    assign v = r_v;
endmodule
```

실행 결과



a의 값이 4'b1000일 때 output값이 11 v값이 1 출력